

Optimization

R. Prerequisites and review: geometry, differential calculus and linear algebra

José Firmino Aguilar Madeira

Why this review?

The optimization methods we will study rest on four tools:

- **Vectors and lines** — each iteration of a method is a step $\mathbf{x}_{k+1} = \mathbf{x}_k + \alpha_k \mathbf{d}_k$ along a line (iterative scheme: see 1.1).
- **Gradient** — when $\nabla f \neq \mathbf{0}$, it gives the direction of greatest increase of f ; $-\nabla f$ is the direction of steepest descent.
- **Hessian and eigenvalues** — help classify stationary points.
- **Convexity** — relates local and global minima (see 1.1).

Vectors, lines and hyperplanes

Angle between two vectors

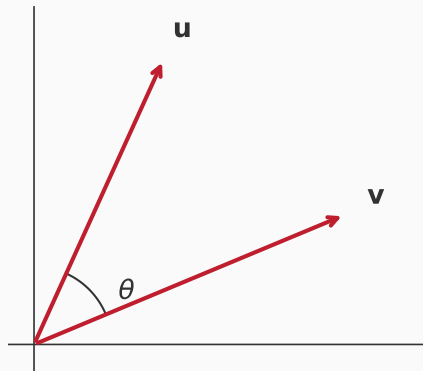
For $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$, both nonzero, the inner product is

$$\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u}^T \mathbf{v} = \sum_{i=1}^n u_i v_i = \|\mathbf{u}\| \|\mathbf{v}\| \cos \theta.$$

Hence the angle $\theta \in [0, \pi]$ between $\mathbf{u}$ and $\mathbf{v}$ is

$$\theta = \arccos\left(\frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{u}\| \|\mathbf{v}\|}\right).$$

- $\langle \mathbf{u}, \mathbf{v} \rangle = 0 \iff \mathbf{u} \perp \mathbf{v}$
- $\langle \mathbf{u}, \mathbf{v} \rangle > 0 \iff \theta < 90^\circ$ (“same direction”)



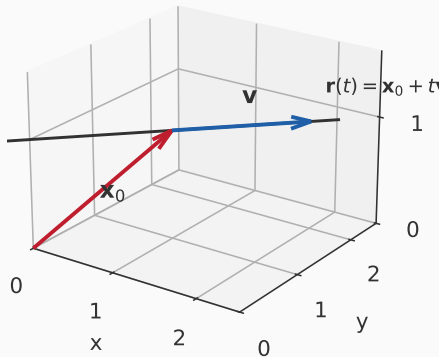
Parametric equation of a line

The line R through $\mathbf{x}_0$ with the direction of $\mathbf{v} \neq \mathbf{0}$ is

$$\mathbf{r}(t) = \mathbf{x}_0 + t \mathbf{v}, \quad t \in \mathbb{R}.$$

- $\mathbf{x}_0$ — initial point
- $\mathbf{v}$ — direction vector
- t — scalar parameter

Each point in space is identified with its position vector.



Parametric equation of a line — examples

Example 1

Line through $\mathbf{x}_0 = (1, 1, 1)^T$ in the direction of $\mathbf{v} = (1, 1, 0)^T$:

$$\mathbf{r}(t) = (1, 1, 1)^T + t(1, 1, 0)^T = (1 + t, 1 + t, 1)^T, \quad t \in \mathbb{R}.$$

Example 2 — from parametric to Cartesian form

Let $\mathbf{r}(t) = (4 - 2t, 3 + 6t)^T$ in $\mathbb{R}^2$. From $x = 4 - 2t$ we get $t = \frac{4 - x}{2}$ and

$$y = 3 + 6 \cdot \frac{4 - x}{2} = 15 - 3x.$$

Line segment

Given two points $\mathbf{u} \neq \mathbf{s}$, the **segment** joining them is

$$\mathbf{r}(t) = \mathbf{u} + t(\mathbf{s} - \mathbf{u}) = (1 - t)\mathbf{u} + t\mathbf{s}, \quad 0 \leq t \leq 1.$$

($t = 0$ gives $\mathbf{u}$, $t = 1$ gives $\mathbf{s}$.)

Example 3

Segment between $(0, 2)$ and $(2, 0)$:

$$\mathbf{r}(t) = (1 - t)(0, 2) + t(2, 0) = (2t, 2 - 2t), \quad 0 \leq t \leq 1.$$

This expression reappears in the definition of a **convex set**.

A hyperplane generalizes the line and the plane to n dimensions:

Space	Equation	Object
$\mathbb{R}^2$	$a_1x_1 + a_2x_2 = b$	line (dimension 1)
$\mathbb{R}^3$	$a_1x_1 + a_2x_2 + a_3x_3 = b$	plane (dimension 2)
$\mathbb{R}^n$	$a_1x_1 + \cdots + a_nx_n = \mathbf{a}^T \mathbf{x} = b, \mathbf{a} \neq \mathbf{0}$	hyperplane (dimension $n - 1$)

- A hyperplane of $\mathbb{R}^n$ is the set $\{\mathbf{x} : \mathbf{a}^T \mathbf{x} = b\}$ with $\mathbf{a} \neq \mathbf{0}$; the vector $\mathbf{a}$ is **normal** to the hyperplane.
- A hyperplane divides $\mathbb{R}^n$ into two half-spaces: $\mathbf{a}^T \mathbf{x} \leq b$ and $\mathbf{a}^T \mathbf{x} \geq b$.
- Linear constraints of an optimization problem define hyperplanes and half-spaces.

Functions of several variables

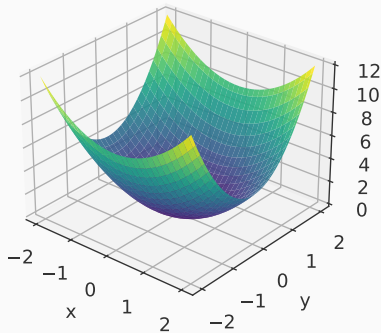
Graph of a function

Given $f: S \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$, the **graph** of f is the subset of $\mathbb{R}^{n+1}$

$$\{(\mathbf{x}, x_{n+1}) \in \mathbb{R}^n \times \mathbb{R} : \mathbf{x} \in S, x_{n+1} = f(\mathbf{x})\}.$$

- $n = 1$: curve $y = f(x)$ in the plane.
- $n = 2$: surface $z = f(x, y)$ in space.

Example. $f(x, y) = x^2 + 2y^2$ is defined on $\mathbb{R}^2$; its graph is a surface in $\mathbb{R}^3$.



Level sets

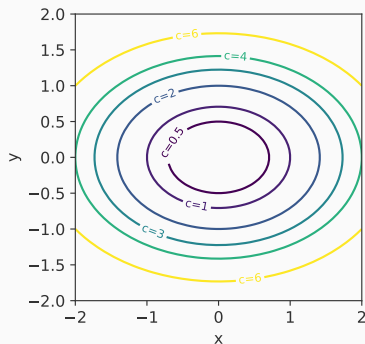
Let $f: D \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ and $c \in \mathbb{R}$. The **level set** c is

$$\{\mathbf{x} \in D : f(\mathbf{x}) = c\},$$

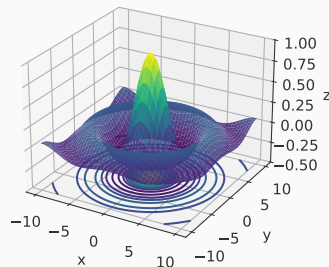
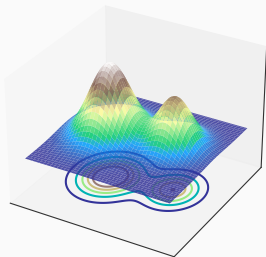
that is, the points of the domain where f equals c . In $\mathbb{R}^2$ it is, in general, a **level curve**; in $\mathbb{R}^3$, a **level surface**.

Example. For $f(x, y) = x^2 + 2y^2$ and $c > 0$, the level curves $x^2 + 2y^2 = c$ are concentric ellipses (c grows outwards). For $c = 0$ the level set reduces to $\{(0, 0)\}$; for $c < 0$ it is empty.

In optimization, level curves are drawn to visualize the path of iterative methods.



Level curves — topographic interpretation



Level curves are the “contour lines” of a topographic map: the projection onto the xy plane of the intersections of the graph with horizontal planes $z = c$.

Directional derivative

We want the rate of change of $z = f(\mathbf{x})$ at the point $\mathbf{a}$ in the direction of a **unit** vector $\mathbf{u}$ ($\|\mathbf{u}\| = 1$).

Definition

Let $f: D \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$, $\mathbf{a}$ interior to D and $\|\mathbf{u}\| = 1$. The directional derivative of f at $\mathbf{a}$ in the direction $\mathbf{u}$ is

$$D_{\mathbf{u}}f(\mathbf{a}) = \lim_{h \rightarrow 0} \frac{f(\mathbf{a} + h\mathbf{u}) - f(\mathbf{a})}{h},$$

if the limit exists.

- The distance between $\mathbf{a}$ and $\mathbf{a} + h\mathbf{u}$ is $|h|$; the quotient is the average rate of change per unit length on the segment from $\mathbf{a}$ to $\mathbf{a} + h\mathbf{u}$.
- It depends on the point $\mathbf{a}$ *and* on the direction $\mathbf{u}$.

Example

Let $f(\mathbf{a})$ be the temperature at the point $\mathbf{a}$ of a room with a radiator and the door open. Moving towards the door, the temperature drops; towards the radiator, it rises. The rate at which it changes depends on where we are and where we are going.

If $\mathbf{u} = \mathbf{e}_i$ (standard basis vector), $D_{\mathbf{e}_i} f(\mathbf{a}) = \frac{\partial f}{\partial x_i}(\mathbf{a})$: partial derivatives are special cases of the directional derivative.

Definition

The gradient of f , denoted ∇f or $\text{grad } f$, is the vector of partial derivatives:

$$\nabla f(\mathbf{x}) = \left(\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}, \dots, \frac{\partial f}{\partial x_n} \right)^T.$$

If f is differentiable, the directional derivative is an inner product:

$$D_{\mathbf{u}}f(\mathbf{x}) = \sum_{j=1}^n \frac{\partial f}{\partial x_j}(\mathbf{x}) u_j = \nabla f(\mathbf{x})^T \mathbf{u} = \|\nabla f(\mathbf{x})\| \cos \theta,$$

where θ is the angle between $\nabla f(\mathbf{x})$ and $\mathbf{u}$.

Lipschitz. f is **L -Lipschitz** on S if $|f(\mathbf{x}) - f(\mathbf{y})| \leq L \|\mathbf{x} - \mathbf{y}\|$ for all $\mathbf{x}, \mathbf{y} \in S$ (Lipschitz gradient: $\|\nabla f(\mathbf{x}) - \nabla f(\mathbf{y})\| \leq L \|\mathbf{x} - \mathbf{y}\|$).

Theorem

If f is differentiable at $\mathbf{x}$ and $\nabla f(\mathbf{x}) \neq \mathbf{0}$, the maximum value of $D_{\mathbf{u}}f(\mathbf{x})$ is $\|\nabla f(\mathbf{x})\|$ and it occurs when $\mathbf{u}$ has the direction and sense of $\nabla f(\mathbf{x})$.

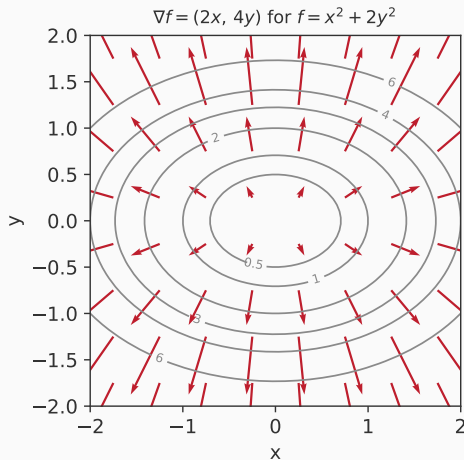
- ∇f points to where f increases fastest; $-\nabla f$ points to where f decreases fastest.
- A direction $\mathbf{d}$ is a **descent** direction at $\mathbf{x}$ if $\nabla f(\mathbf{x})^T \mathbf{d} < 0$ (angle with the gradient greater than 90°).
- It is the basis of the steepest descent method.

Gradient and level curves in $\mathbb{R}^2$

If $\nabla f(x_0, y_0) \neq \mathbf{0}$, the gradient, besides giving the direction of greatest increase, is **perpendicular to the level curve** $f(x, y) = c$ through (x_0, y_0) .

Along the level curve f is constant, so the directional derivative in the tangent direction is zero: $\nabla f^T \mathbf{u}_{\text{tang}} = 0$.

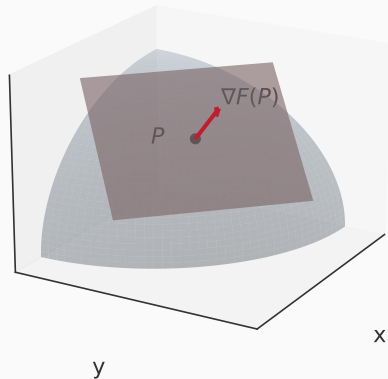
Example. $f = x^2 + 2y^2$, $\nabla f = (2x, 4y)^T$: the arrows cross the ellipses at right angles and point outwards (where c grows).



Gradient and level surfaces in $\mathbb{R}^3$

If $\nabla F(P) \neq \mathbf{0}$, the gradient $\nabla F(x_0, y_0, z_0)$ is **perpendicular to the tangent plane** to the level surface $F(x, y, z) = c$ through $P = (x_0, y_0, z_0)$.

If $\mathbf{r}(t)$ is a curve on the surface with $\mathbf{r}(t_0) = P$, then $F(\mathbf{r}(t)) = c$ and, differentiating, $\nabla F(P)^T \mathbf{r}'(t_0) = 0$: the gradient is orthogonal to all tangent vectors.



Tangent plane and normal line to a surface

Let $F(x, y, z) = c$ and $P = (x_0, y_0, z_0)$ a point of the surface with $\nabla F(P) \neq \mathbf{0}$.

- **Tangent plane** — vectors starting at P that are orthogonal to $\nabla F(P)$:

$$\nabla F(P)^T (x - x_0, y - y_0, z - z_0) = 0.$$

- **Normal line** — through P with the direction of $\nabla F(P)$:

$$(x - x_0, y - y_0, z - z_0) = s \nabla F(P), \quad s \in \mathbb{R},$$

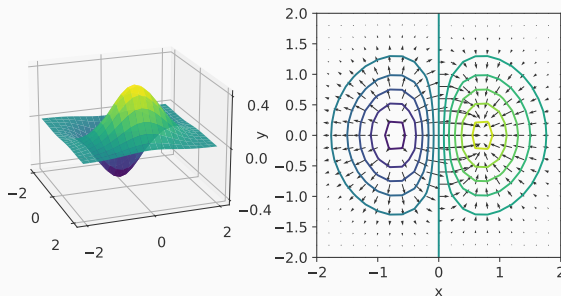
or, in symmetric equations (when the partial derivatives are nonzero),

$$\frac{x - x_0}{F_x(P)} = \frac{y - y_0}{F_y(P)} = \frac{z - z_0}{F_z(P)}.$$

Visualizing gradients in MATLAB

```
[X,Y] = meshgrid(-2:.2:2);  
Z = X.*exp(-X.^2 - Y.^2);  
[DX,DY] = gradient(Z,.2,.2);  
subplot(1,2,1); surf(X,Y,Z)  
subplot(1,2,2); contour(X,Y,Z)  
hold on; quiver(X,Y,DX,DY)
```

Surface $z = x e^{-x^2 - y^2}$ and the gradient field over the level curves.



Hessian matrix and eigenvalues

Hessian matrix

Let $f: A \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ be of class C^2 in a neighbourhood of $\mathbf{a} \in A$. The **Hessian matrix** of f at $\mathbf{a}$ (also $\nabla^2 f(\mathbf{a})$; in the methods, $H_k = \nabla^2 f(\mathbf{x}_k)$) is

$$Hf(\mathbf{a}) = \left[\frac{\partial^2 f}{\partial x_i \partial x_j}(\mathbf{a}) \right]_{i,j=1}^n = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & \cdots & \frac{\partial^2 f}{\partial x_2 \partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \frac{\partial^2 f}{\partial x_n \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{bmatrix}.$$

Since $f \in C^2$, it is a **real symmetric** $n \times n$ matrix (Schwarz). We will see next that this guarantees real eigenvalues, and that their sign helps classify stationary points.

Definition

Given $A \in \mathbb{R}^{n \times n}$, a scalar λ and a vector $\mathbf{x} \neq \mathbf{0}$ such that

$$A\mathbf{x} = \lambda\mathbf{x}$$

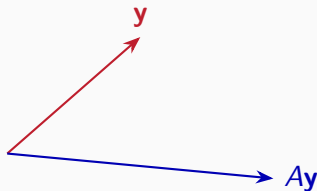
are called, respectively, an **eigenvalue** and an **eigenvector** of A .

- λ may be complex even when A is real (but not if A is symmetric).
- **Spectrum:** $\lambda(A) = \{\lambda_1, \dots, \lambda_n\}$, the set of eigenvalues.
- **Spectral radius:** $\rho(A) = \max\{|\lambda_1|, \dots, |\lambda_n|\}$.

Geometric interpretation



- $Ax = \lambda x$
- Ax has the direction of x
- the length and sense of Ax depend on λ



- $Ay \neq \lambda y$ for every λ
- Ay does not have the direction of y
- y is not an eigenvector of A

Computing eigenvalues and eigenvectors

$$A\mathbf{x} = \lambda\mathbf{x} \iff A\mathbf{x} - \lambda\mathbf{x} = \mathbf{0} \iff (A - \lambda I)\mathbf{x} = \mathbf{0}.$$

We want **nontrivial** solutions ($\mathbf{x} \neq \mathbf{0}$) of the homogeneous system, which requires the coefficient matrix to be singular:

$$\det(A - \lambda I) = 0.$$

- $\det(A - \lambda I)$ is the **characteristic polynomial**, of degree n in λ ; its roots are the eigenvalues.
- For each λ_i , the eigenvectors are the nonzero solutions of $(A - \lambda_i I)\mathbf{x} = \mathbf{0}$.

Example 1

$$A = \begin{bmatrix} 2 & 1 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad \det(A - \lambda I) = (2 - \lambda)^2(-1 - \lambda)(1 - \lambda).$$

Eigenvalues: 2 (algebraic multiplicity 2), -1 and 1 (multiplicity 1).

Example 2

$$A = \begin{bmatrix} 2 & 2 & -1 \\ 1 & -1 & 0 \\ 3 & 1 & -1 \end{bmatrix}, \quad \det(A - \lambda I) = 2\lambda - \lambda^3 = -\lambda(\lambda - \sqrt{2})(\lambda + \sqrt{2}).$$

Eigenvalues: $\lambda_1 = 0$, $\lambda_2 = \sqrt{2}$, $\lambda_3 = -\sqrt{2}$.

Positive definite matrix

Definition

A symmetric matrix $A \in \mathbb{R}^{n \times n}$ is **positive definite** if

$$\mathbf{x}^T A \mathbf{x} > 0 \quad \text{for every } \mathbf{x} \neq \mathbf{0},$$

and **positive semidefinite** if $\mathbf{x}^T A \mathbf{x} \geq 0$ for every $\mathbf{x}$.

- A is positive definite $\iff$ all eigenvalues are positive.
- A is positive semidefinite $\iff$ all eigenvalues are ≥ 0 .

Example

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}: \quad \det(A - \lambda I) = \begin{vmatrix} 2 - \lambda & 1 \\ 1 & 2 - \lambda \end{vmatrix} = \lambda^2 - 4\lambda + 3 = 0 \Rightarrow \lambda_1 = 3, \lambda_2 = 1. \text{ Hence}$$

A is positive definite.

Sylvester's criterion

Let A be symmetric. A is positive definite if and only if all **leading principal minors** are positive:

$$\det[a_{11}] > 0, \quad \det \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} > 0, \quad \det \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} > 0, \quad \dots, \quad \det A > 0.$$

With Δ_k the leading minor of order k : A is **negative definite** if and only if $(-1)^k \Delta_k > 0$, $k = 1, \dots, n$.

Caution

For positive *semidefiniteness* it is not enough to look at the leading minors: *all* principal minors (not only the leading ones) must be ≥ 0 . Example: $\begin{bmatrix} 0 & 0 \\ 0 & -1 \end{bmatrix}$ has leading minors 0 and 0, but is not positive semidefinite ($\lambda = -1$).

Classification of stationary points

Let $\mathbf{a}$ be a stationary point ($\nabla f(\mathbf{a}) = \mathbf{0}$) and let $\lambda_{\min}$, $\lambda_{\max}$ be the smallest and largest eigenvalues of $Hf(\mathbf{a})$. A matrix A is **negative (semi)definite** if $-A$ is positive (semi)definite; **indefinite** if $\mathbf{x}^T A \mathbf{x}$ takes both signs. The second derivative test:

Eigenvalues of $Hf(\mathbf{a})$	$Hf(\mathbf{a})$	Conclusion
all positive	positive definite	$\mathbf{a}$ is a strict local minimum
all negative	negative definite	$\mathbf{a}$ is a strict local maximum
$\lambda_{\min} = 0$, $\lambda_{\max} > 0$	positive semidefinite	$\mathbf{a}$ is not a maximum
$\lambda_{\min} < 0$, $\lambda_{\max} = 0$	negative semidefinite	$\mathbf{a}$ is not a minimum
$\lambda_{\min} < 0$, $\lambda_{\max} > 0$	indefinite	$\mathbf{a}$ is a saddle point
all zero	—	inconclusive

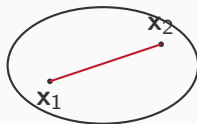
In the semidefinite and zero cases the test does not decide; higher-order terms must be analysed. (Strict local minimum: see 1.1.)

Convexity

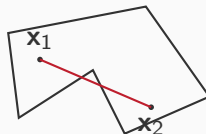
Definition 1

A set $S \subseteq \mathbb{R}^n$ is **convex** if, for any $\mathbf{x}_1, \mathbf{x}_2 \in S$, the segment joining them is contained in S :

$$(1 - t)\mathbf{x}_1 + t\mathbf{x}_2 \in S \quad \text{for every } t \in [0, 1].$$



convex



not convex

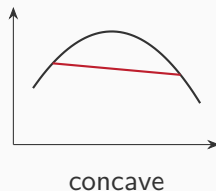
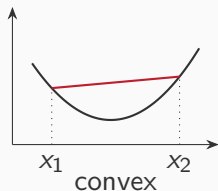
Convex function

Definition 2

$f: S \rightarrow \mathbb{R}$, with S convex, is **convex** if the segment joining any two points of the graph never lies below the graph:

$$f((1-t)\mathbf{x}_1 + t\mathbf{x}_2) \leq (1-t)f(\mathbf{x}_1) + tf(\mathbf{x}_2), \quad t \in [0, 1].$$

It is **concave** if the inequality holds with $\geq$ (that is, $-f$ is convex).



Convexity and optimization

Theorem 1

In a convex optimization problem (convex objective and feasible set; see 1.1), **every local minimum is global**.

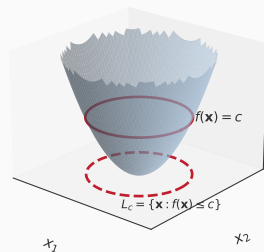
Theorem 2

If f is convex, the sublevel set $L_c = \{\mathbf{x} : f(\mathbf{x}) \leq c\}$ is convex for every $c \in \mathbb{R}$.

Theorem 3

The intersection of convex sets is convex.

Consequence: $\{g_j(\mathbf{x}) \geq 0, h_\ell(\mathbf{x}) = 0\}$ with g_j concave and h_ℓ affine is convex (course convention $g_j \geq 0$: see 1.1).



Characterizations of convexity

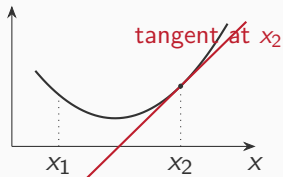
Theorem 4

Let f be of class C^2 on an open convex set S . The following are equivalent:

- (a) f is convex on S ;
- (b) $f(\mathbf{x}_1) \geq f(\mathbf{x}_2) + \nabla f(\mathbf{x}_2)^T(\mathbf{x}_1 - \mathbf{x}_2)$ for all $\mathbf{x}_1, \mathbf{x}_2 \in S$ (the graph lies above every tangent plane);
- (c) the Hessian $Hf(\mathbf{x})$ is **positive semidefinite** for every $\mathbf{x} \in S$.

- Hf positive definite on $S \Rightarrow f$ *strictly* convex (the converse is false: $f(x) = x^4$ is strictly convex but $f''(0) = 0$).
- f convex $\Rightarrow \nabla f(\mathbf{x}^*) = \mathbf{0}$ is a *necessary and sufficient* condition for a global minimum (unconstrained; $\mathbf{x}^*$ interior to S).

Characterizations of convexity (cont.)



Theorem 4(b): the graph lies above the tangent.

Theorem 5

A quadratic form $q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$ with A positive semidefinite is convex.

Theorem 6

A linear combination with nonnegative coefficients of convex functions is convex.

Theorem 7

f is convex on S if and only if, for all $\mathbf{x} \in S$ and $\mathbf{d}$, the function of one variable $\varphi(\alpha) = f(\mathbf{x} + \alpha \mathbf{d})$ (the one in 2.1) is convex on the interval $\{\alpha : \mathbf{x} + \alpha \mathbf{d} \in S\}$.

KEY TAKEAWAYS — R

- **Line** $\mathbf{r}(t) = \mathbf{x}_0 + t\mathbf{v}$: each iteration of a method is a step along a line.
- **Gradient** ∇f : direction of greatest increase, perpendicular to the level curves; $\nabla f^T \mathbf{d} < 0$ defines a descent direction.
- **Hessian** Hf : symmetric; the sign of the eigenvalues helps classify stationary points (semidefinite cases: inconclusive).
- **Positive definite** (A symmetric): $\mathbf{x}^T A \mathbf{x} > 0 \iff \text{eigenvalues} > 0 \iff \text{leading minors} > 0$.
- **Convexity** ($f \in C^2$ on an open convex set): Hessian positive semidefinite $\iff f$ convex $\Rightarrow$ a local minimum is global.

References Full list: *Course bibliography*, at the end of the course PDF.

- Nocedal, J., Wright, S. J., *Numerical Optimization*, 2nd ed., Springer, 2006 — Ch. 2.
- Dennis, J. E., Jr., Schnabel, R. B., *Numerical Methods for Unconstrained Optimization and Nonlinear Equations*, SIAM (Classics in Applied Mathematics 16), 1996.

Resources

- MATLAB: `meshgrid`, `gradient`, `surf`, `contour`, `quiver` (visualizing gradients).